

Mohammad Ninad Mahmud Nobo

Bangladesh University of Engineering and Technology (BUET) | CSE

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SUMMARY

B.Sc. graduate in Computer Science and Engineering from BUET with experience in building intelligent systems at the intersection of large language models, software automation, and applied machine learning. Hands-on experience developing AI-driven applications across backend and mobile platforms with a strong focus on software engineering and scalable system design. Research interests include LLM-based software testing, AI evaluation, and medical AI, with experience in developing practical AI-powered solutions.

EDUCATION

Bangladesh University of Engineering and Technology (BUET) 2022 – 2026
Bachelor of Science in Computer Science and Engineering (CSE) CGPA: 3.60 / 4.00
Relevant Coursework: Data Structures and Algorithms, Digital Logic Design, Database Systems, Operating Systems, Computer Architecture, Computer Networks, Computer Graphics, Algorithm Engineering, Artificial Intelligence, Machine Learning
Research Interests: LLM Systems, Software Testing Automation, Applied Machine Learning, Medical AI, Computer Networks

RESEARCH EXPERIENCE

- **AutoTestGenX: Multi-Agent LLM Framework for End-to-End Web Testing** [[GitHub](#)]
Python • LLM Engineering • Multi-Agent Systems • Web Testing • Browser Automation 2025 – 2026
 - Developed a multi-agent LLM framework for automated web test generation and execution from natural-language requirements.
 - Built benchmark datasets and evaluated generated test suites against engineer-validated ground-truth test suites.
 - Achieved 84.0% test scenario coverage and 90% error detection, outperforming zero-shot and few-shot prompting approaches.
- **MedCAR: Conflict-Aware Medical Reasoning System (Under Review)** [[GitHub](#)]
Python • Deep Learning • Medical AI 2026
 - Built a multi-model system for chest X-ray analysis integrating heterogeneous AI tools.
 - Developed a conflict resolution pipeline to reconcile contradictory outputs using semantic reasoning and confidence calibration.
 - Designed trust-aware decision framework with uncertainty-based abstention for reliable predictions.
- **Bengali-Loop: Community Benchmarks for Long-Form Bangla ASR and Speaker Diarization** [[arXiv: 2602.14291](#)]
Python • Speech Processing • Machine Learning 2026
 - Contributed to building a large-scale benchmark dataset for long-form Bangla ASR and speaker diarization in a multi-institution collaboration, developing data collection and annotation pipelines for real-world speech data.
 - Designed subtitle extraction and annotation workflows enabling evaluation with WER and DER.

PROJECTS

- **MindTrace: AI-Powered Dementia Care Application** [[GitHub](#) | [Feature](#) | [Infrastructure](#)]
Kotlin • Spring Boot • Spring AI • PostgreSQL • Redis • Firebase • Docker • Azure 2025
 - Engineered an AI-driven mobile-backend system for dementia care with an Android frontend and scalable Spring Boot architecture, enabling real-time caregiver workflows and seamless system integration.
 - Designed accessibility-first UI/UX for cognitively impaired users, focusing on clarity, simplified interactions, and usability to enhance engagement and reduce cognitive load.
- **Gemma VetCare: AI-Assisted Farming Support System** [[GitHub](#) | [Feature](#)]
Kotlin • Android Studio • Spring Boot • Spring AI • MongoDB 2025
 - Built an AI-assisted Android application for veterinary decision support, integrating backend services to deliver real-time recommendations for diagnosis and treatment workflows.
 - Designed RESTful APIs optimized for low-connectivity environments with efficient request handling and streamlined data flow.
- **Compiler Construction (Flex, Bison, 8086 Codegen)** [[GitHub](#)]
C++17 • Flex • Bison/Yacc • Linux 2024
 - Designed and implemented a full compiler pipeline for a C-like language, encompassing lexical analysis, parsing, semantic analysis, and intermediate code generation.
 - Generated optimized 8086 assembly code with symbol table management and basic optimizations.
- **Computer Graphics Pipeline Implementation** [[GitHub](#)]
C++17 • OpenGL/GLUT • Rasterization • Ray Tracing 2025
 - Implemented a complete graphics pipeline including transformations, clipping, rasterization with Z-buffering, and ray tracing.
 - Developed OpenGL-based rendering demos with camera systems supporting 6-DOF control and real-time interaction.

TECHNICAL SKILLS

Languages: C/C++, Python, Java, Kotlin, JavaScript, SQL
Frontend: React, Embedded JavaScript (EJS), HTML, CSS3
Backend & APIs: Spring Boot, Spring AI, Node.js, Express.js, REST APIs, Secure API Design
AI/ML: PyTorch, TensorFlow, Scikit-learn, Large Language Models (LLMs), Prompt Engineering, Multi-Agent Systems
Mobile Development: Android, Kotlin, Android Studio
Databases: PostgreSQL, MongoDB, Redis
DevOps & Tools: Linux, Git, Docker, Docker Compose, GitHub Actions, Firebase

ADDITIONAL INFORMATION

Spoken Languages: Bangla (Native), English (Fluent), Hindi & Urdu (Conversational), Arabic (Reading)
Programming Profiles: [NeetCode: mninadmnobo](#) [LeetCode: mninadmnobo](#)
Research Profiles: [ORCID: 0009-0006-2781-6693](#) [ResearchGate: Mohammad Ninad Mahmud Nobo](#)